Varunprasaath R. Selvaraju, Ph.D.

(+1) 678-207-9914 | varunprasaath@gmail.com | linkedin.com/in/varunprasaath-rs |

PROFESSIONAL SUMMARY

PhD chemist/materials scientist with 6+ years in organic/materials chemistry, reaction and process optimization, and data-rich analytical characterization (NMR, LC-MS/GC-MS, FT-IR, GPC/SEC, TGA/DSC, DOE/SPC). Currently leading DOE-guided process development for bio-based polymer systems, translating experimental data into reproducible process windows and scale-up specifications.

CORE COMPETENCIES

Process Chemistry | Reaction Optimization | DOE/SPC | Analytical Readouts | Organic Synthesis | Data-Rich Experimentation | Process Documentation

WORK EXPERIENCE

University of Colorado Boulder
July 2025 - Present

Postdoctoral Researcher

- Lead DOE-guided chemical process development for bio-based polymer systems, defining critical process parameters, material-quality readouts, and reproducible scale-up specifications.

University of Colorado Boulder
July 2021 - July 2025

Graduate Researcher

- Designed, synthesized, and characterized organic electrochromic materials, connecting molecular structure, purity, stability, and performance across 15+ variants with NMR/GPC, MS, spectroscopy, thermal analysis, and electrochemistry.

Georgia Institute of Technology
Aug 2019 - July 2021

Graduate Researcher

- Drove process development for CVD-based molecular thin-film coatings, optimizing surface preparation and deposition conditions to improve reproducibility and optical performance.

ETH Zurich
May 2017 - July 2017

Research Assistant

- Optimized CVD polymer thin-film deposition under flow using DOE and in-situ QCM-D monitoring to quantify growth kinetics and process controls.

University of Colorado Boulder
2021 - 2025

Laboratory Operations Manager

- Managed characterization/process-development lab operations for a 15-member group; implemented calibration, SOP, and safety systems that improved instrument uptime by 30%.

Georgia Tech
2019-2021

Teaching Assistant, Organic Chemistry I & Lab

- Mentored 120+ undergraduates; developed supplemental materials improving exam performance by 15%.

KEY PROJECTS

Bio-Based Polymer Scale-Up Process `Current`

Built DOE-guided process windows, analytical readouts, and material-quality specifications for reproducible polymer scale-up.

DOE, SPC, process variables, analytical characterization

Electrochromic Molecular Design `Chemistry of Materials 2025`

Designed and evaluated molecular variants through synthesis, purification, characterization, and property-driven iteration.

Organic synthesis, NMR, MS, FT-IR, UV-Vis-NIR, CV

AFOSR Defense Thin-Film Coatings `DoD Funded`

Converted early-stage coating concepts into controlled CVD process conditions and reproducible thin-film outputs.

CVD, QCM-D, surface prep, process optimization

EDUCATION

Ph.D. in Chemistry - University of Colorado Boulder 2025

Dissertation: Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials
Advisor: Prof. Seth Marder
(Transferred from Georgia Institute of Technology)

B.S. (Research), Major: Chemistry; Minor: Materials Science - Indian Institute of Science (IISc) 2019

INSPIRE Scholarship for Higher Education (Top 1% nationally)

CERTIFICATIONS

Statistical Process Control (SPC) - Coursera, 2026

TECHNICAL SKILLS

Process & Reaction Development: DOE, process optimization, scale-up specifications, reaction/workup analysis, reproducibility studies, SOPs
Analytical Characterization: NMR, LC-MS, GC-MS, GPC/SEC, FT-IR, UV-Vis-NIR, TGA, DSC, QCM-D, cyclic voltammetry
Chemistry: Organic synthesis, small-molecule materials, polymers, purification, structure-property analysis
Statistical & Data: SPC, JMP (basic), Python (basic), Excel, structured experimental records, technical reports
Materials: Bio-based polymers, electrochromic molecules, thin films, CVD coatings, surface functionalization

JOURNAL PUBLICATIONS

1. Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Narrow-Band Electrochromism of a Ferrocene-Substituted Rhodamine B." Chemistry of Materials, 2025, 37, 5558-5568.
2. Mandal, J.; Ramasamy Selvaraju, V.; Yan, W.; Divandari, M.; Spencer, N. D.; Dubner, M. "In Situ Monitoring of SI-ATRP Throughout Multiple Reinitiations Under Flow by Means of a Quartz Crystal Microbalance." RSC Advances, 2018, 8, 20048-20055.
3. Ramasamy Selvaraju, V.; Venta, R.; Rajapaksha, I.; Zhang, J.; Barlow, S.; Salman, S.; Scott, C.; Marder, S. "Beyond Rhodamine B: Investigating Alternative Xanthene Cores for Modular Electrochromic Systems." (Manuscript under preparation).

CONFERENCE PRESENTATIONS

Modular Design and Electrochemical Properties of Xanthene-Based Electrochromic Materials, ACS Fall National Meeting, Denver, CO, August 2024.
Electrochemical and Spectroscopic Properties of Substituted Rhodamine Derivatives, Gerischer-Lewerenz Electrochemistry Symposium, Fort Collins, CO, August 2024.
Decoupling Redox and Color for High-Contrast Electrochromics, Excited State Processes Conference, Santa Fe, NM, June 2023.
Electrochromism: Decoupling Color and Redox in Rhodamine-Based Systems, Rocky Mountain Solid State Chemistry Workshop, Boulder, CO, January 2023.

HONORS & AWARDS

Plastic Sustainability Innovation in Material Award | INSPIRE Scholarship (Top 1%, India, 2015) | 97.12 Percentile - IIT JEE Mains (2015) | All India Rank 144 - NEST (2014)